

Hierarchical Models of the Visual System

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Synonyms

Hierarchical architectures; Deep learning architectures; Simple-to-complex hierarchies; Hubel & Wiesel model; Large-scale models of the visual system.

Definition

Hierarchical models of the visual system are neural networks with a layered topology: In these networks, the receptive fields of units (i.e., the region of the visual space that units respond to) at one level of the hierarchy are constructed by combining inputs from units at a lower level. After a few processing stages, small receptive fields tuned to simple stimuli get combined to form larger receptive fields tuned to more complex stimuli. Such anatomical and functional hierarchical architecture is a hallmark of the organization of the visual system.

Since the pioneering work of Hubel and Wiesel (1962), a variety of hierarchical models have been described from relatively small-scale models of the primary visual cortex to very large-scale (system-level) models of object and action recognition, which account for processing in large portions of the visual field and entire visual streams. The term ‘model of the visual system’ is generally reserved for architectures that are constrained in some way by the anatomy and the physiology of the visual system (with various degrees of realism). Convolutional networks are closely related connectionist networks with a similar architecture that have been used in multiple real-world machine learning problems including speech and music classification.

Detailed description

The processing of shape information in the ventral stream of the visual cortex follows a series of stages, starting from the retina, through the Lateral Geniculate Nucleus (LGN) of the thalamus to primary visual cortex (V1) and extrastriate visual areas, V2, V4 and the inferotemporal (IT) cortex. In turn IT provides a major source of input to prefrontal cortex (PFC) involved in linking

perception to memory and action (see DiCarlo et al., 2012, for a recent review but also *Limitations* section below for further discussion). As one progresses along the ventral stream visual hierarchy, neurons become selective for increasingly complex stimuli – from simple oriented bars and edges in early visual area V1 to moderately complex features in intermediate areas (such as combination of orientations) and complex objects and faces in higher visual areas such as IT. In parallel to this increase in the complexity of the preferred stimulus, the invariance properties of neurons also increases with neurons gradually becoming more and more tolerant with respect to the exact position and scale of the stimulus within their receptive fields. As a result of this increase in invariance properties, the receptive field size of neurons increases, from about one degree or less in V1 to several degrees in IT.

Hierarchical models of the visual system have a long history starting with Marko and Giebel (1970)’s homogeneous multi-layered architecture and later Fukushima (1980)’s Neocognitron. One of the key principles in the Neocognitron and other modern hierarchical models originates from the pioneering physiological studies and models of Hubel and Wiesel (1962). In these networks, the receptive fields of units at one level of the hierarchy are constructed by combining inputs from units at a lower level. After a few processing stages, small receptive fields tuned to simple stimuli get combined to form larger receptive fields tuned to more complex stimuli.

Several hierarchical models of the ventral stream of the visual system have been described since the Neocognitron to account for the organization and the neurophysiology of the ventral stream of the visual cortex. These models can be coarsely divided into conceptual proposals (Biederman, 1987; Perrett and Oram, 1993; Hochstein and Ahissar, 2002) and neurobiological models (e.g., ?Mel, 1997; Riesenhuber and Poggio, 1999; Ullman et al., 2002; Thorpe, 2002; Amit and Mascaró, 2003; Wersing and Koerner, 2003; Serre et al., 2007; Masquelier and Thorpe, 2007; Grossberg et al., 2011a,b; O’Reilly et al., 2013). Similar hierarchical models have also been proposed to explain motion processing in the dorsal stream of the visual cortex (e.g., Simoncelli and Heeger, 1998; Grossberg et al., 1999; Perrone and Thiele, 2002; Giese and Poggio, 2003; Rust et al., 2006; Jhuang et al., 2007; Pack and Born, 2008; Mineault et al., 2012).

Somewhat independently, convolutional networks and other deep learning architectures have been developed in computer vision (LeCun et al., 1998). These neural networks do not mimic the organization of the visual cortex in detail, but biology is often cited as a source of inspiration. While these models are not, strictly speaking, models of the visual system, their impressive success in multiple visual recognition tasks (Krizhevsky et al., 2012), offer supporting evidence for hierarchical models of the visual system.

Hierarchical models of the primary visual cortex

Hubel and Wiesel (1962) first described two functional classes of cortical cells: Simple cells respond best to oriented stimuli (e.g, bars, edges, gratings) at one particular orientation, position and phase (i.e., white bar on a black background

Figure 1: **Hubel & Wiesel model.** (A) Receptive field (RF) of a simple cell obtained by selectively pooling over afferent (center-surround) cells aligned along a preferred axis of orientation (horizontal shown here). (B) At the next stage, a complex cell RF can be obtained by selectively pooling over afferent simple cells with the same preferred orientation (horizontal). Shown here is a complex cell RF obtained by pooling over position to build tolerance to translation of the preferred stimulus but a more complete model of a complex cell would also include pooling over simple cells tuned to slightly different spatial frequency and phases (Rust et al., 2005; Chen et al., 2007). Modified from Hubel and Wiesel (1962).

or a dark bar on a white background) within their relatively small receptive fields. Complex cells, on the other hand, while also selective for bars, tend to have larger receptive fields (about twice as large) and exhibit some tolerance with respect to the exact position of the stimulus within their receptive fields. Complex cells are also invariant to contrast reversal, i.e., the same cell responds to a white bar on a black background or the opposite.

Figure 1 illustrates a plausible neural circuit proposed by Hubel and Wiesel (1962) to explain the receptive field organization of these two functional classes of cells. Simple-cell-like receptive fields can be obtained by pooling the activity of a small set of cells tuned to spots of lights with a center-surround organization (as observed in ganglion cells in the LGN and layer IV of the striate cortex) aligned along a preferred axis of orientation (Figure 1A).

Similarly, at the next stage, position tolerance at the complex cell level, could be obtained by pooling over afferent simple cells (from the level below) with the same preferred (horizontal) orientation but slightly different positions (Figure 1B). While the precise circuits underlying the invariance of complex cells are still debated (Chance et al., 2000), the coarse circuitry and underly-

ing pooling mechanisms postulated by Hubel & Wiesel over fifty years ago are now relatively well established (Rust et al., 2005; Chen et al., 2007). This has recently lead to the development of a number of computational models of the primary visual cortex (Traub et al., 2005; Zhu et al., 2010; Antolík and Bednar, 2011; Bednar, 2012), which account for the processing of one or a few cortical hypercolumns (see below) at the level of biophysically realistic circuits.

Hierarchical models of the visual system

Hierarchical models of the visual system typically extend these two classes of simple and complex cells from striate to extrastriate areas and come in many different forms: They differ in terms of their specific wiring and corresponding parameterizations as well as the mathematical operations that are implemented. However, common to all these models is an underlying basic architecture corresponding to multiple stages of processing as shown for the HMAX, a representative hierarchical model of visual processing (Riesenhuber and Poggio, 1999; Serre et al., 2007), on Figure 2.

A general wiring diagram of the HMAX and other related hierarchical models of the visual cortex is shown on Figure 3. Units at stage $k + 1$ pool selectively over afferent units from the previous stage k within a local neighborhood (shown in red). In general, pooling may occur over multiple dimensions of the afferent units (e.g., position, scale, orientation, etc). Pooling over multiple locations (as shown in stage k on Figure 3) leads to an increase in the receptive field size of the units at the next stage (compare the receptive field size of a unit at stage k shown in yellow with that of a unit at a higher stage $k + 1$ (shown in red).

For instance, a computational instantiation of the Hubel & Wiesel hierarchical model of the primary visual cortex corresponds to two processing stages (a more detailed model would include an additional processing stage corresponding to center-surround cells – not shown here for simplicity). Simple cells in layer $k = 1$ receive their inputs directly from pixel intensities in the previous layer $k = 0$ (see yellow receptive field in Figure 3). Complex cells in layer $k = 2$ pool over afferent simple cells at the same orientation over a local neighborhood (shown is a 3×3 neighborhood). These types of circuits have lead to several models of the primary visual cortex that have focused on explaining in reasonably good biophysical details the tuning properties of individual cells such as orientation, motion or stereo disparity, (see Landy and Movshon, 1991, for review).

Figure 2: **Sketch of the Hmax hierarchical model of visual processing:** Acronyms: V1, V2 and V4 correspond to primary, second and fourth visual areas, PIT and AIT to posterior and anterior inferotemporal areas, respectively (tentative mapping with areas of the visual cortex shown in color, some areas of the parietal cortex and dorsal streams not shown). The model relies on two types of computations: A max-like operation (shown in dash circles) over similar features at different position and scale to gradually build tolerance to position and scale and a bell-shaped tuning operation (shown in plain circle) over multiple features to increase the complexity of the underlying representation. Since it was originally developed Riesenhuber and Poggio (1999), the model has been able to explain a number of new experimental data (Serre et al., 2007). This includes data that were not used to derive or fit model parameters. The model seems to be qualitatively and quantitatively consistent with (and in some cases actually predicts) several properties of subpopulations of cells in V1, V4, IT, and PFC as well as fMRI and psychophysical data (see Serre and Poggio, 2010, for a recent review).

Figure 3: Hierarchical models of the visual system are characterized by multiple stages of processing whereby units in one stage (shown as squares) pool the response of units from the previous stage. Individual stages (also called layers) contain multiple *feature maps* organized in terms of both spatial location and scale. A hypercolumn contains all possible features from all feature maps for that location. Hence each stage can be thought of as containing hypercolumns replicated at all positions and scales.

In recent years, because of the increasing amount of computing power available, the scale of models of visual processing has increased with models now encompassing large portions of the visual field and entire streams of visual processing (see Poggio and Serre, 2013, for a review). Alternating between multiple layers of simple units and complex units leads to an architecture that is able to achieve a difficult trade-off between selectivity and invariance: Along the hierarchy, at each “simple unit” stage, units become tuned to features of increasing complexity (e.g., from single oriented bars, to combinations of oriented bars to form corners and features of intermediate complexities) by combining afferents (complex units) with different selectivities (e.g., units tuned to edges at different orientations). Conversely, at each “complex unit” stage, units become increasingly invariant to 2D transformations (position and scale) by combining afferents (simple units) with the same selectivity (e.g., a vertical bar) but slightly different positions and scales.

While recent work has suggested that simple and complex cells may represent the ends of a continuum instead of two-discrete classes of neurons (see Ringach (2004) for a discussion), this dichotomy is probably not critical for hierarchical models of the visual system. Indeed, some recent models do not distinguish between simple and complex cell pooling (O’Reilly et al., 2013).

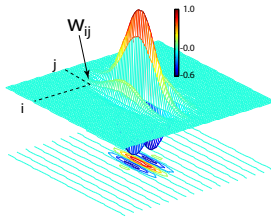


Figure 4: Linear filter model of a simple cell using Gabor functions (see text for details).

Columnar organization

Units in hierarchical models of the visual cortex are typically organized in columns and/or feature maps. A hypercolumn (shown in blue on Figure 3) corresponds to a population of units tuned to a basic set of features (e.g., units spanning the full range of possible orientations or directions of motion, etc) in models of the primary visual cortex (Hubel and Wiesel, 1962). These hypercolumns are then replicated at all positions in the visual field and multiple scales. An alternative perspective is to think of processing stages in terms of feature maps. Typically, maps correspond to retinotopically organized population of units tuned to the same feature (e.g., specific motion direction, orientation, binocular disparity, etc) but at multiple positions (tiling the visual space) and/or multiple scales. This model is often referred to as the ice-cube model and more complex models of columnar organization have been proposed since (e.g., to account for pinwheel centers Von der Malsburg (1973)), hierarchical models of the visual system typically follow the ice-cube model for its simple implementation.

In addition to the feedforward (bottom-up) connections, which correspond to projections from processing stage k to $k^* > k$, units can also be connected via lateral (horizontal) connections (both short-range connections within a hypercolumn and long range between hypercolumns at different retinal locations) or feedback (top-down) connections from processing stage k to $k^* < k$ (e.g., (Wallis and Rolls, 1997; O'Reilly et al., 2013)).

Neuronal operations

Different hierarchical models of the visual system differ in terms of the operation that they implement. Examples of operations include both linear and non-linear pooling stages. Linear pooling can be described by the following operation:

$$y = \sum_{i \in I} w_i x_i = \mathbf{w} \cdot \mathbf{x}, \quad (1)$$

where y is a scalar that corresponds to the activity of the pooling unit (over a set of afferent units I), $\mathbf{w} = (w_1, \dots, w_k, \dots, w_n)$ is the vector of synaptic weights (often referred to as a linear “filter”, the neuron linear receptive field

or “projective field”) for the corresponding input vector $\mathbf{x} = (x_1, \dots, x_k, \dots, x_n)$. The pool of units I is typically organized in terms of hypercolumns / feature maps (see above). For instance, in models of the primary visual cortex, the synaptic weights of a simple cell tuned to spatial orientation can be thought of as a 2D filter parameterized by the weight matrix $W = \{w_{ij}\}$, w_{ij} corresponds to the weight of the unit at location (i, j) within the receptive field. Here we consider two dimensional receptive fields for simplicity but in the general case receptive fields can have many dimensions including time, feature types, eye dominance, etc.

A good 2D mathematical model of simple cells’ receptive fields is the Gabor function (Jones and Palmer, 1987), which is given by the following equation:

$$w_{ij} = \exp\left(-\frac{(\hat{i}^2 + \gamma^2 \hat{j}^2)}{2\sigma^2}\right) \times \cos\left(\frac{2\pi \hat{i}}{\lambda}\right) \quad (2)$$

s.t. $\hat{i} = i \cos \theta + j \sin \theta$ and $\hat{j} = -i \sin \theta + j \cos \theta$.

The five parameters, i.e., the orientation θ , the aspect ratio γ , the effective width σ , the phase ϕ and the wavelength λ , determine the properties of the spatial receptive field of the corresponding model simple cell (see Figure 4 for an example of a 2D Gabor function). Also there exists a straightforward 3D extension of this parameterization to include a temporal receptive field (see Dayan and Abbott, 2001).

A simple extension of the linear model is the linear-nonlinear (LN) cascade model whereby the unit response y is passed to a non-linear scalar function f with non-negative output (Rieke et al., 1997):

$$y = f\left(\sum_{i \in I} w_i x_i\right) \sim \sum_{i \in I} w_i^* g(x_i), \quad (3)$$

Popular choices for the function f include rectification functions, exponential functions, or the logistic and the hyperbolic function). The LN model has been shown to account for a host of experimental data (Rieke et al., 1997) and it has been shown that in many cases, biophysically more realistic, spiking neuron models can be reduced to a simple LN cascade (Ostojic and Brunel, 2011). The LN cascade (linear filter followed by the hyperbolic function) has also been the main pooling operation used in standard connectionist networks.

Extensions of the LN cascade include the addition of a normalization stage, in which the response y of the neuron is divided by a common factor that typically includes the summed activity of a pool of neurons:

$$y = \frac{\sum_{i \in I} w_i f(x_i)}{k + \sum_{j \in J} g(x_j)}, \quad (4)$$

where $k \ll 1$ is a constant to avoid zero-division. The pool of neurons J used for normalization may correspond to the same pool of neurons I which shape the

(classical) receptive field or may extend beyond to account for extra-classical receptive field effects (Series et al., 2003). Normalization circuits were originally proposed to explain the contrast response of cells in the primary visual cortex and are now thought to operate throughout the visual system, and in many other sensory modalities and brain regions (see Carandini and Heeger, 2012, for a recent review).

For instance, in the HMAX model shown on Figure 2, two types of operations have been assumed: A bell-shaped tuning functions for simple cells and a max-like operation at the level of complex cells (Riesenhuber and Poggio, 1999). Interestingly both operations can be approximated via a specific form of Equation 4. Mathematically, a bell-shaped tuning function and a softmax, take essentially the same general form, that is:

$$y = \frac{\sum_{i \in I} w_i x_i^p}{k + \left(\sum_{i \in I} x_i^q \right)^r}, \quad (5)$$

where p , q and r represent the static nonlinearities in the underlying neural circuit. Such nonlinearity may correspond to different regimes on the $f - I$ curve of the presynaptic neurons such that different operating ranges provide different degrees of nonlinearities (from near-linearity to steep non-linearity). An extra sigmoid transfer function on the output $g(y) = 1/(1 + \exp^{\alpha(y-\beta)})$ controls the sharpness of the unit response. By adjusting these non-linearities, Eq. 5 can approximate better a max or a tuning function (see Kouh and Poggio, 2008, for details).

Why hierarchies?

The most studied visual function is probably object recognition, which reflects our ability to assign a label or meaning to an image of an object irrespective of the precise size, position, illumination or context and clutter. The main computational problem in object recognition is achieving invariance while preserving selectivity (Riesenhuber and Poggio, 1999); cells found in IT are typically tuned to views of complex objects such as faces (Tsao and Livingstone, 2008) – they discharge strongly to a face but very little or not at all to other objects. A hallmark of these cells is the robustness of their responses in the face of stimulus transformations such as scale and position changes. This finding presents an interesting question: How could these cells respond differently to similar stimuli (for instance, two different faces) that activate the retinal photoreceptors in similar ways, but respond consistently to scaled and translated versions of the preferred stimulus, which produce very different activation patterns on the retina? It has been postulated that the goal of the ventral stream of the visual cortex is to achieve an optimal tradeoff between selectivity and invariance

via a hierarchy of processing stages whereby neurons at higher and higher levels exhibit an increasing degree of invariance to image transformations such as translations and scale changes (Riesenhuber and Poggio, 1999).

Now, why hierarchies? Several hypotheses have been described to explain the hierarchical modularity of our visual system ranging from a stability argument (i.e., faster adaptation or evolution of the system in response to changing environmental conditions) to minimization of wiring cost in addition to other evolutionary arguments (see Meunier et al., 2010, for review). An alternative hypothesis – for models in the Hubel & Wiesel spirit – is that the hierarchy may provide a solution to the invariance-selectivity trade-off problem by decomposing a complex task such as invariant object recognition in a hierarchy of simpler ones (at each stage of processing). Hierarchical organization in cortex is not limited to the visual pathways, and thus a more general explanation may be needed. Interestingly, from the point of view of classical learning theory (Poggio and Smale, 2003), there is no need for architectures with more than three layers.

So, why hierarchies? There may be reasons of efficiency, such as the efficient use of computational resources. For instance, the lowest levels of the hierarchy may represent a dictionary of features that can be shared across multiple classification tasks (Geman, 1999). There may also be the more fundamental issue of sample complexity, the number of training examples required for good generalization (see Serre and Poggio, 2010, for a discussion). An obvious difference between the best classifiers derived from learning theory and human learning is in fact the number of examples required in tasks such as object recognition. Statistical learning theory shows that the complexity of the hypothesis space sets the requirement for the number of samples needed for learning. If a task – like a visual recognition task – can be decomposed into low-complexity learning tasks for each layer of a hierarchical learning machine, then each layer may require only a small number of training examples (Poggio and Smale, 2003). Neuroscience suggests that what humans can learn may be represented by hierarchies that are locally simple. Thus, our ability to learn from just a few examples, and its limitations, may be related to the hierarchical architecture of cortex.

Limitations

To-date most existing hierarchical models of visual processing both from the perspective of biological and machine vision are instances of feedforward models (but see O’Reilly et al., 2013). These models have been useful to explore the power of fixed hierarchical organization as originally suggested by Hubel and Wiesel (1962). These models assume that our core object recognition capability proceeds through a cascade of hierarchically organized areas of the visual cortex with computations at each successive stage being largely feedforward (Riesenhuber and Poggio, 1999; DiCarlo et al., 2012). They have led, for instance, to algorithms competitive with the best computer vision systems (see Serre and Poggio, 2010, for a review). Their limitations, however, are becoming increasingly obvious. Not only top-down effects are key to normal, everyday vision,

but back-projections are also likely to be a key part of what cortex is computing and how. Human observers can essentially answer an infinite number of questions about an image (one could in fact imagine a Turing test of vision). Thus, a major question for modeling visual cortex revolves around the role of back-projections and the related fact that vision is more than object recognition and requires interpreting and parsing visual scenes (as opposed to simply finding out whether a specific object is present in the visual scene or not).

In addition, while the overall hierarchical organization of the visual cortex is now well established (Felleman and Van Essen, 1991), the parallel between the anatomical and functional hierarchy is, however, looser than one might expect. While the trend is, from lower to higher visual areas, for neurons' receptive fields to become increasingly large and tuned to increasingly complex preferred stimuli, there remains a very broad distribution of tuning and receptive field sizes in all areas of the visual hierarchy. For instance, IT, which is commonly assumed to have solved the problem of invariant recognition (DiCarlo et al., 2012), also contains neurons with relatively small receptive fields and tuned to relatively simple visual features such as simple orientations (Desimone et al., 1984). A close comparison of shape representation between V1, V2 and V4 also demonstrated a complex pattern of shape selectivity with significant deviation from strict hierarchical organization with some cells in V1 exhibiting more complex tuning than some cells in V4 (Hegde and Van Essen, 2007). Furthermore, beside the visual cortical hierarchy, there exist additional subcortical pathways (including cortico-thalamo-cortical loops). Hence, the anatomical hierarchy should be taken as an idealization and cannot be taken as a strict flowchart of visual information (Hegd  and Felleman, 2007).

Cross-References

Feedforward network, Recurrent network, Deep learning network

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